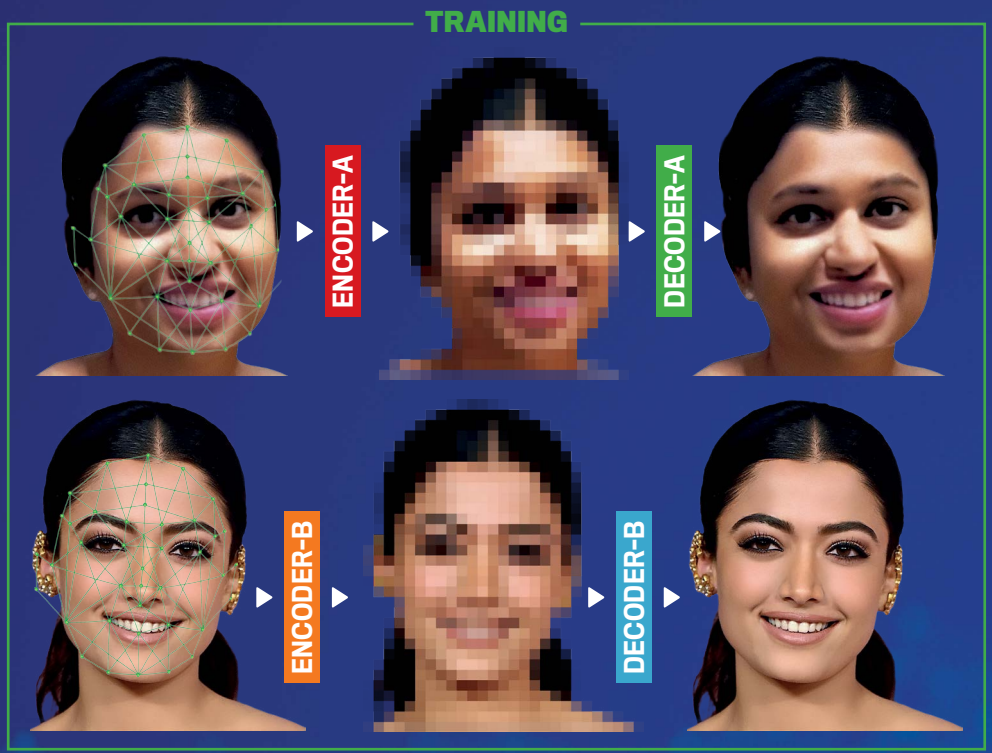
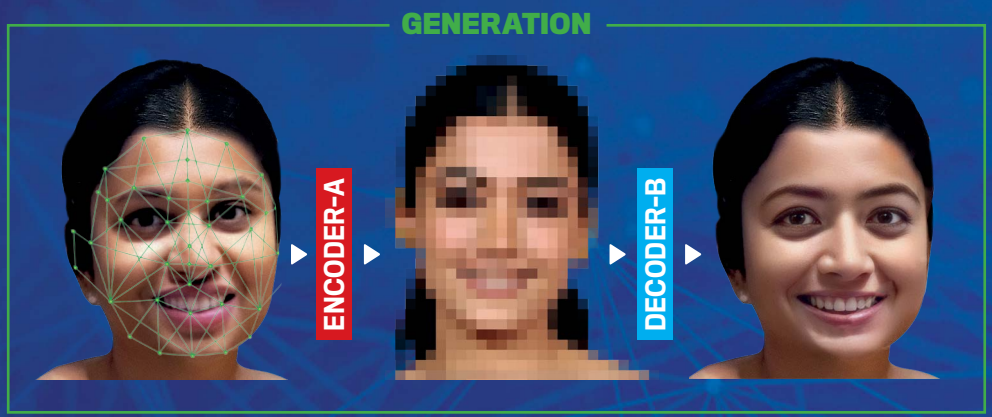
HOW DEEPFAKES ARE MADE

IN 2017, A REDDIT HANDLE CALLED 'DEEPFAKE' STARTED POSTING DOCTORED CLIPS THAT SWAPPED THE FACES OF CELEBRITIES WITH PORN ACTORS. THAT'S HOW THE TERM WAS COINED–THOUGH VIDEO AND IMAGE MANIPULATION IS SOMETHING WHOSE BOUNDARIES KEEP EXPANDING WITH EMERGING TECH

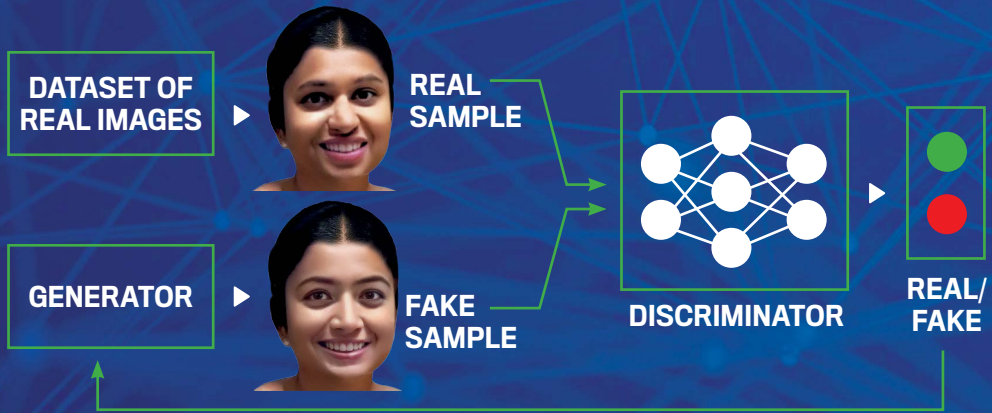
Graphic by **NILANJAN DAS & TANMOY CHAKRABORTY**



1 At the heart of a deepfake is an autoencoder. An encoder takes an image and reduces it to a small number of variables. The decoder then takes this compact representation and expands it into the original image. Next, the training algorithm feeds another image to the network and the whole process repeats. It can take thousands of iterations to produce an autoencoder that does a good job of reproducing its own input



2 Once you've trained a pair of autoencoders, you switch decoders. You encode a photo of, say, A and B, but then use the decoder of A in the decoding of B. The result is a reconstructed photo of B, but with the same head pose and expression as in the original photo of A



3 Another type of machine learning, known as Generative Adversarial Networks (GANs), is added to this mix. It detects and improves flaws in the deepfake within multiple rounds, making it harder for deepfake detectors to decode them